

MUHAMMAD HASSAN

Computer Science Student | Software Engineering & IT

CONTACT INFORMATION

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PROFESSIONAL SUMMARY

Final-year Computer Science (BSCS) student with a strong foundation in data structures, algorithms, object-oriented programming, and database systems. Completed a Web Development internship with excellent performance (90%, Grade A), and proven ability to design and build practical applications using Python, C++, JavaScript, and Flask, demonstrated through a portfolio of academic and personal projects spanning algorithm visualization, systems programming, and database design. Quick learner, detail-oriented, and eager to contribute to entry-level Software Engineering, Computer Science, or IT roles where I can apply strong problem-solving skills and a passion for clean, maintainable code.

EDUCATION

Bachelor of Science in Computer Science (BSCS) 2022 — 2026

Pak-Austria Fachhochschule: Institute of Applied Sciences and Technology

- Relevant coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Operating Systems, Computer Networks, Software Engineering

Intermediate (Pre-Engineering) 2020 — 2022

KIPS College, Wah Cantt, Punjab, Pakistan — Mathematics, Physics, Computer Science

SKILLS

Programming Languages: Python, C++, JavaScript

Web Technologies: HTML, CSS, Flask

Tools & Platforms: Git, GitHub, Miniconda, MongoDB

Concepts: Object-Oriented Programming, Data Structures & Algorithms, Database Design (SQL/ERD), Operating Systems Concepts, Digital Logic Design

Soft Skills: Problem Solving, Analytical Thinking, Team Collaboration, Time Management

PROJECTS

Airline Reservation System

Technologies: C++, OOP, Data Structures

- Designed and implemented an object-oriented airline booking system handling flights, seat allocation, and passenger records using inheritance and encapsulation.

DSA Sorting Visualizer

Technologies: JavaScript, HTML5, Algorithms

- Built an interactive web visualizer animating Bubble, Merge, and Quick Sort algorithms step by step to aid algorithm comprehension.

CPU Page Replacement Visualizer

Technologies: C++, Operating Systems, Algorithms

- Developed an Operating Systems project visualizing FIFO, LRU, and Optimal page replacement algorithms and comparing page-fault rates in real time.

Disaster Management System

Technologies: SQL, Database Design, ERD

- Modeled and managed disaster response data — resources, shelters, and emergency reports — through a relational database schema.

Shifa Medicines — Hospital Management System

Technologies: Python, Miniconda, OOP

- Built an in-memory hospital management system handling patient records and medicine inventory using core OOP principles.

Water Level Indicator

Technologies: Physics, Electronics, Hardware

- Created a sensor-based circuit to detect and display water levels, alerting when tanks reach critical levels.

Automatic Light Detector (Sun Tracker)

Technologies: Digital Logic Design, Circuit Design, Hardware

- Designed an LDR-sensor and logic-circuit system to automatically orient a solar panel toward the strongest light source.

EXPERIENCE

Web Development Intern

July 2025 — August 2025

M Tech Production and Marketing

- Completed a 1-month, hands-on Web Development internship, gaining practical experience in real-world development workflows.
- Achieved a 90% (Grade A) performance score, fulfilling all internship requirements with dedication and creativity.

CERTIFICATIONS

- Web Development Internship Certificate of Completion — M Tech Production and Marketing (August 2025), Grade A (90%)
- MongoDB Database by Giraffe Academy — Cursa (December 2025)